INCH-POUND
A-A-59650A
22 May 2006
SUPERSEDING
A-A-59650
31 August 2001

COMMERCIAL ITEM DESCRIPTION

BEARINGS, ROLLER, TAPERED, SINGLE ROW, NORMAL ANGLE, FLANGED CUP (TYPE TSF)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes government acquisition requirements for complete (cone with rollers and cup) single row, tapered, roller bearings with normal angle, flanged cup (type TSF) for general-purpose use. These bearings are not intended for use in special precision applications such as on aircraft, precision ordnance or submarine equipment.

2. CLASSIFICATION. The roller bearings shall be of one type (TSF) and classified by the size codes listed in table I. The column headings in table I refer to bearing characteristics defined in figure 1.

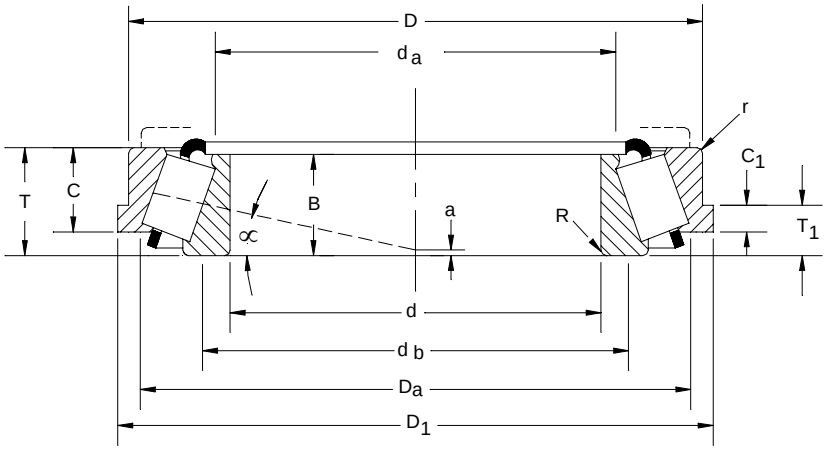


FIGURE 1. Bearing configuration.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3110

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TABLE I. Size codes and dimensions.*

Size code	Part 1/ number	d	D	T	T1	B	C	R 2/	D1	C1	d b	d a	D a	K factor	Basic dynamic load ratings (lb.)	a 3/
		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Max. shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
001	A4050-A4138-B	0.5000	1.3775	0.4330	0.1823	0.4326	0.3437	0.05	1.4985	0.0930	0.73	0.67	1.32	1.29	2100	-0.10
002	A4059-A4138-B	0.5906	1.3775	0.4330	0.1823	0.4326	0.3437	0.03	1.4985	0.0930	0.77	0.75	1.32	1.29	2100	-0.10
003	17580-17520-B	0.6250	1.6875	0.6563	0.2500	0.6563	0.5315	0.06	1.8085	0.1250	0.91	0.83	1.59	1.76	5100	-0.23
004	05062-05185-B	0.6250	1.8504	0.5662	0.2377	0.5662	0.4375	0.06	2.0024	0.1090	0.93	0.83	1.75	1.64	4400	-0.16
005	A6075-A6157-B	0.7500	1.5745	0.4730	0.1910	0.4391	0.3750	0.04	1.6955	0.0930	0.94	0.91	1.50	1.11	2200	-0.06
006	09067-09195-A-B	0.7500	1.9380	0.7100	0.2755	0.7500	0.5625	0.05	2.0898	0.1280	1.00	0.94	1.83	2.20	6900	-0.29
007	1380-1328-B	0.8750	2.0625	0.7625	0.3560	0.7940	0.5625	0.06	2.2147	0.1560	1.16	1.06	1.97	2.00	7800	-0.30
008	1779-1729-B	0.9375	2.2400	0.7625	0.2935	0.7810	0.6250	0.03	2.3920	0.1560	1.16	1.12	2.09	1.90	7300	-0.27
009	3659-3620-B	0.9375	2.4375	1.1250	0.3750	1.1975	0.9375	0.09	2.6209	0.1875	1.40	1.24	2.32	2.07	14200	-0.47
010	07100-07196-B	1.0000	1.9687	0.5313	0.2657	0.5614	0.3750	0.04	2.1209	0.1094	1.20	1.16	1.93	1.45	4800	-0.11
011	07100-07204-B	1.0000	2.0470	0.5910	0.2000	0.5614	0.5000	0.04	2.1990	0.1090	1.20	1.16	1.97	1.45	4800	-0.11
012	1718-1729-B	1.0000	2.2400	0.7625	0.2935	0.7810	0.6250	0.03	2.3920	0.1560	1.20	1.18	2.09	1.90	7300	-0.27
013	15101-15250-B	1.0000	2.5000	0.8125	0.3437	0.8125	0.6250	0.03	2.6522	0.1562	1.28	1.24	2.36	1.67	8200	-0.23
014	15580-15520-B	1.0625	2.2500	0.6875	0.3125	0.6875	0.5313	0.14	2.4022	0.1563	1.52	1.26	2.17	1.69	7500	-0.20
015	15590-15520-B	1.1250	2.2500	0.6875	0.3125	0.6875	0.5313	0.14	2.4022	0.1563	1.56	1.32	2.17	1.69	7500	-0.20
016	1985-1931-B	1.1250	2.3750	0.7813	0.3125	0.7620	0.6250	0.03	2.5270	0.1562	1.34	1.32	2.24	1.77	8300	-0.23
017	02474-02420-B	1.1250	2.6875	0.8750	0.3437	0.8750	0.6875	0.03	2.8397	0.1562	1.44	1.42	2.56	1.40	10400	-0.20
018	3198-3120-B	1.1250	2.8593	1.1875	0.4375	1.1810	0.9375	0.05	3.0433	0.1875	1.54	1.46	2.72	1.76	15200	-0.40
019	17119-17244-B	1.1875	2.4409	0.6300	0.2075	0.6522	0.5625	0.06	2.5930	0.1400	1.46	1.36	2.32	1.53	7000	-0.14

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		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Max. shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
020	02476-02420-B	1.2500	2.6875	0.8750	0.3437	0.8750	0.6875	0.03	2.8397	0.1562	1.54	1.52	2.32	1.40	10400	-0.20
021	02475-02420-B	1.2500	2.6875	0.8750	0.3437	0.8750	0.6875	0.14	2.8397	0.1562	1.75	1.52	2.56	1.40	10400	-0.20
022	2580-2523-B	1.2500	2.7500	0.9375	0.3435	0.9983	0.7500	0.03	2.9020	0.1560	1.52	1.48	2.60	2.14	13500	-0.34
023	2581-2523-B	1.3125	2.7500	0.9375	0.3435	0.9983	0.7500	0.03	2.9020	0.1560	1.56	1.54	2.60	2.14	13500	-0.34
024	2585-2523-B	1.3125	2.7500	0.9375	0.3435	0.9983	0.7500	0.14	2.9020	0.1560	1.77	1.54	2.60	2.14	13500	-0.34
025	3196-3120-B	1.3125	2.8593	1.1875	0.4375	1.1810	0.9375	0.14	3.0433	0.1875	1.85	1.59	2.72	1.76	15200	-0.40
026	31594-31520-B	1.3750	3.0000	1.1563	0.4063	1.1250	0.9375	0.06	3.1836	0.1875	1.181	1.71	2.91	1.45	15200	-0.30
027	31597-31520-B	1.4375	3.0000	1.1563	0.4063	1.1250	0.9375	0.14	3.1836	0.1875	2.01	1.75	2.91	1.45	15200	-0.30
028	13889-13836-B	1.5000	2.5625	0.5000	0.2340	0.4688	0.3750	0.06	2.6835	0.1090	1.77	1.67	2.48	1.69	4500	-0.03
029	19150-19268-B	1.5000	2.6875	0.6250	0.2968	0.6504	0.4688	0.06	2.8397	0.1406	1.77	1.69	2.64	1.31	8300	-0.06
030	19150-19283-B	1.5000	2.8346	0.6700	0.2475	0.6504	0.5625	0.06	2.9865	0.1400	1.77	1.69	2.68	1.31	8300	-0.06
031	16150-16484-B	1.5000	2.8440	0.8125	0.3438	0.8125	0.6250	0.14	2.9960	0.1563	1.95	1.69	2.72	1.45	9200	-0.16
032	3490-3420-B	1.5000	3.1250	1.1563	0.4063	1.1721	0.9375	0.14	3.3090	0.1875	2.05	1.80	2.99	1.60	16800	-0.34
033	28150-28315-B	1.5000	3.1496	0.8270	0.0583	0.8244	0.6250	0.06	3.3015	0.1563	1.79	1.71	2.87	1.45	16000	-0.19
034	3382-3320-B	1.5625	3.1562	1.1563	0.4063	1.1965	0.9375	0.14	3.3396	0.1875	2.05	1.79	3.03	2.14	18500	-0.43
035	11162-11300-B	1.6250	3.0000	0.7090	0.2871	0.6844	0.5625	0.06	3.1836	0.1406	1.93	1.83	2.87	1.20	7800	-0.03
036	26882-26822-B	1.6250	3.1250	0.9375	0.3438	0.1000	0.7500	0.14	3.2772	0.1563	2.13	1.85	2.99	1.83	14700	-0.29
037	3877-3820-B	1.6250	3.3750	1.1875	0.4375	1.1875	.9375	0.14	3.5270	0.1875	2.24	1.98	3.27	1.45	2.0000	-0.32
038	LL103049-LL103010-B	1.7500	2.8125	0.5000	0.2350	0.5000	0.3750	0.06	2.9225	0.1100	2.01	1.91	2.72	1.91	5900	-0.05

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	T	T1	B	C	R 2/	D1	C1	d b	d a	D a	K factor	Basic dynamic load ratings (lb.)	a 3/
		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Max. shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
039	L102849-L102810-B	1.7500	2.8750	0.7188	0.2500	0.7188	0.5938	0.06	3.0000	0.1250	2.01	1.93	2.80	1.83	9300	-0.15
040	3578-3525-B	1.7500	3.4375	1.1875	0.4370	1.2160	0.9375	0.14	3.6215	0.1870	2.24	2.01	3.23	1.91	18100	-0.40
041	49175-49368-B	1.7500	3.6875	1.2500	4.6885	1.2500	1.0000	0.14	3.9073	0.2188	2.32	2.09	3.50	1.62	22000	-0.36
042	438-432-B	1.7500	3.7500	1.0938	0.4376	1.1772	0.8750	0.14	3.9640	0.2188	2.24	2.01	3.43	2.05	20500	-0.36
043	18690-18620-B	1.8125	3.1250	0.6875	0.2968	0.6875	0.5313	0.11	3.3086	0.1406	2.20	2.01	3.03	1.56	8500	-0.08
044	25592-25521-B	1.8125	3.2700	0.9375	0.3438	1.0000	0.7500	0.14	3.4220	0.1563	2.28	2.05	3.15	1.74	14500	-0.25
045	359-S-354-B	1.8125	3.3465	0.8125	0.3125	0.8540	0.6875	0.09	3.5299	0.1875	2.17	2.01	3.23	1.91	13300	-0.19
046	2984-2924-B	1.8125	3.3465	1.0000	0.3750	1.0082	0.8125	0.14	3.5340	0.1875	2.28	2.05	3.23	1.69	15100	-0.25
047	436-432-B	1.8125	3.7500	1.0938	0.4376	1.1772	0.8750	0.14	3.9640	0.2188	2.32	2.05	3.43	2.05	20500	-0.36
048	65390-65320-B	1.9375	4.5000	1.7500	0.6563	1.7500	1.3750	0.14	4.7772	0.2813	2.76	2.36	4.21	1.36	35700	-0.49
049	L305649-L305610-B	2.0000	3.1875	0.7188	0.2813	1.7188	0.5625	0.06	3.3085	0.1250	2.28	2.20	3.07	1.64	9900	-0.10
050	18790-18720-B	2.0000	3.3465	0.6875	0.2968	0.6875	0.5313	0.14	3.4870	0.1406	2.44	2.20	3.23	1.44	8500	-0.03
051	368A-362-B	2.0000	3.5433	0.7874	0.3499	0.8750	0.6250	0.14	3.7268	0.1875	2.44	2.20	3.39	1.83	14000	-0.17
052	3780-3720-B	2.0000	3.6718	1.1875	0.4375	1.1930	0.9375	0.14	3.8558	0.1875	2.52	2.28	3.54	1.73	19700	-0.32
053	49585-49520-B	2.0000	4.0000	1.2500	0.4688	1.2500	1.0000	0.14	4.2148	0.2188	2.60	2.32	3.86	1.46	21300	-0.28
054	529-522-B	2.0000	4.0000	1.3750	0.5625	1.4200	1.0625	0.03	4.2460	0.2500	2.32	2.28	3.82	2.05	26400	-0.50
055	529X-522-B	2.0000	4.0000	1.3750	0.5625	1.4200	1.0625	0.14	4.2460	0.2500	2.56	2.28	3.82	2.05	26400	-0.50
056	65395-65320-B	2.0000	4.5000	1.7500	0.6563	1.7500	1.3750	0.14	4.7772	0.2813	2.83	2.36	4.21	1.36	35700	-0.49
057	28584-28521-B	2.0625	3.6250	0.9688	0.3437	1.0000	0.7813	0.14	3.7772	0.1582	2.56	2.28	3.50	1.55	16000	-0.19

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		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Max. shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
058	33895-33820-B	2.1250	3.6718	1.0938	0.4063	1.1250	0.8750	0.06	3.8558	0.1875	2.48	2.36	3.58	1.77	20900	-0.30
059	539-532-B	2.1250	4.3750	1.5000	0.5625	1.4550	1.1875	0.14	4.6210	0.2500	2.68	2.40	3.94	1.97	27600	-0.48
060	621-612-B	2.1250	4.7500	1.6250	0.6562	1.6250	1.2500	0.14	5.0272	0.2812	2.76	2.48	4.33	1.86	33200	-0.55
061	L507949-L507910-B	2.2500	3.4375	0.7188	0.2813	0.7188	0.5625	0.06	3.5625	0.1250	2.56	2.44	3.35	1.50	9400	-0.03
062	387A-382-B	2.2500	3.8125	0.8268	0.3125	0.8640	0.7018	0.14	3.9960	0.1875	2.72	2.44	3.70	1.65	14800	-0.12
063	28682-28622-B	2.2500	3.8437	0.9688	0.3592	0.9688	0.7656	0.14	3.9960	0.1560	2.76	2.48	3.70	1.45	16800	-0.13
064	469-453-B	2.2500	4.2500	1.0938	0.4376	1.1542	0.8750	0.14	4.4640	0.2188	2.76	2.48	3.94	1.74	21800	-0.28
065	623-612-B	2.2500	4.7500	1.6250	0.6562	1.6250	1.2500	0.14	5.0272	0.2812	2.83	2.60	4.33	1.86	33200	-0.55
066	555S-552-B	2.2500	4.8750	1.5000	0.5625	1.4440	1.1875	0.14	5.1210	0.2500	2.87	2.64	4.57	1.69	30700	-0.37
067	28985-28921-B	2.3750	3.9370	1.0000	0.3750	1.0000	0.7813	0.14	4.0930	0.1563	2.87	2.64	3.86	1.37	17200	-0.10
068	65237-65500-B	2.3750	5.0000	1.7500	0.6563	1.7500	1.3750	0.14	5.2772	0.2813	3.23	2.79	4.72	1.20	33300	-0.37
069	39250-39412-B	2.5000	4.1250	0.8438	0.4063	0.8661	0.6250	0.08	4.3084	0.1875	2.87	2.72	4.02	1.51	15800	-0.06
070	29585-29520-B	2.5000	4.2500	1.0000	0.4063	1.0000	0.7500	0.14	4.4022	0.1563	3.03	2.80	4.13	1.27	14200	-0.03
071	390A-394A-B	2.5000	4.3307	0.8661	0.3120	0.8660	0.7411	0.06	4.5147	0.1870	2.87	2.76	4.17	1.45	16200	-0.03
072	29585-29521-B	2.5000	4.3307	1.0000	0.4063	1.0000	0.7500	0.14	4.4803	0.1563	3.03	2.80	4.13	1.27	14200	-0.03
073	3984-3920-B	2.5000	4.8125	1.1875	0.4375	1.1830	0.9375	0.14	4.6210	0.1875	3.03	2.80	4.25	1.45	22400	-0.18
074	5595-5535-B	2.5938	4.8125	1.7188	0.5313	1.7230	1.4375	0.14	5.0625	0.2500	3.27	3.03	4.65	1.27	14200	-0.48
075	29590-29520-B	2.6250	4.2500	1.0000	0.4063	1.0000	0.7500	0.14	4.4022	0.1563	3.15	2.87	4.13	1.27	14200	-0.03
076	395A-394A-B	2.6250	4.3307	0.8661	0.3120	0.8660	0.7411	0.03	4.5147	0.1870	2.87	2.87	4.17	1.45	16200	-0.03
077	395-S-394A-B	2.6250	4.3307	0.8661	0.3120	0.8660	0.7411	0.14	4.5147	0.1870	3.11	2.87	4.17	1.45	16200	-0.03

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078	29590-29521-B	2.6250	4.3307	1.0000	0.4063	1.0000	0.7500	0.14	4.4803	0.1563	3.15	2.87	4.13	1.27	14200	-0.03
079	3984-3920-B	2.6250	4.4375	1.1875	0.4375	1.1830	0.9375	0.14	4.6210	0.1875	3.15	2.91	4.25	1.45	22400	-0.18
080	39590-39520-B	2.6250	4.4375	1.1875	0.4688	1.1875	0.9375	0.14	4.6523	0.2188	3.15	2.91	4.33	1.72	27000	-0.26
081	641-632-B	2.6250	5.3750	1.6250	0.6560	1.6250	1.2500	0.14	5.6520	0.2810	3.27	3.03	4.92	1.61	37400	-0.44
082	399A-394A-B	2.6875	4.3307	0.8661	0.3120	0.8660	0.7411	0.09	4.5147	0.1870	3.07	2.91	4.17	1.45	16200	-0.03
083	560-S-552-B	2.6875	4.8750	1.5000	0.5625	1.1440	1.1875	0.14	5.1210	0.2500	3.27	2.99	4.57	1.69	30.700	-0.37
084	H715343-H715310-B	2.6875	5.5000	1.8125	0.6875	1.8125	1.4375	0.14	6.0000	0.3125	3.54	3.31	5.31	1.24	43000	-0.34
085	29675-29620-B	2.7500	4.4375	1.0000	0.4063	1.0000	0.7500	0.06	4.5900	0.1563	3.15	3.03	4.33	1.20	18000	-0.04
086	LM613449-LM613410-B	2.7500	4.4375	0.8750	0.4375	0.8660	0.6250	0.06	4.6875	0.1875	3.07	2.99	4.33	1.40	16400	-0.00
087	482-472-B	2.7500	4.7244	1.1730	0.4368	1.1420	0.9542	0.14	4.9384	0.2180	3.27	3.03	4.53	1.52	23100	-0.16
088	H414249-H414210-B	2.8125	5.3750	1.6250	0.6550	1.6250	1.2500	0.14	5.6550	0.2800	3.50	3.27	5.12	1.62	41800	-0.43
089	644-632-B	2.8125	5.3750	1.6250	0.6560	1.6250	1.2500	0.14	5.6520	0.2810	3.43	3.19	4.92	1.61	37400	-0.44
090	33287-33462-B	2.8750	4.6250	1.1875	0.4375	1.1875	0.9375	0.14	4.8084	0.1875	3.43	3.15	4.49	1.34	22300	-0.11
091	567-563-B	2.8750	5.0000	1.4375	0.5625	1.4240	1.1250	0.14	5.2460	0.2500	3.46	3.19	4.76	1.69	30700	-0.32
092	42688-42620-B	3.0000	5.0000	1.1875	0.5313	1.2205	0.8750	0.25	5.2460	0.2188	3.78	3.31	4.88	1.39	26100	-0.11
094	6461A-6420-B	3.0000	5.8750	2.1250	0.6875	2.1350	1.7500	0.38	6.1835	0.3125	4.25	3.52	5.51	1.61	55100	-0.59
095	LM814849-LM814810-B	3.0625	4.6250	1.0000	0.4375	1.0000	0.7500	0.14	4.8084	0.1875	3.58	3.35	4.57	1.15	19200	-0.09
096	34306-34478-B	3.0625	4.7812	0.9688	0.4688	0.9060	0.6875	0.14	4.9648	0.1875	3.54	3.31	4.65	1.15	19200	-0.06

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	T	T1	B	C	R 2/	D1	C1	d b	d a	D a	K factor	Basic dynamic load ratings (lb.)	a 3/
		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Max. shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
097	42690-42620-B	3.0625	5.0000	1.1875	0.5313	1.2205	0.8750	0.14	5.2460	0.2188	3.58	3.35	4.88	1.39	26100	-0.11
098	H715348-H715310-B	3.0625	5.5000	1.8125	0.6875	1.8125	1.4375	0.14	6.0000	0.3125	3.86	3.48	5.31	1.24	43000	-0.34
099	L116149-L116110-B	3.2500	4.5625	0.8125	0.3125	0.8438	0.6563	0.06	4.7187	0.1563	3.54	3.46	4.45	1.90	8700	-0.05
100	47686-47620-B	3.2500	5.2500	1.3125	0.5000	1.3125	1.0313	0.14	5.4650	0.2188	3.82	3.54	5.12	1.44	29100	-0.17
101	580-572-B	3.2500	5.5115	1.4375	0.5625	1.4212	1.1250	0.14	5.7575	0.2500	3.86	3.58	5.28	1.45	29800	-0.21
102	663-652-B	3.2500	6.0000	1.6250	0.6562	1.6250	1.2500	0.14	6.2772	0.2812	3.90	3.62	5.55	1.43	39600	-0.31
103	497-493-B	3.3750	5.3750	1.1875	0.5305	1.1720	0.8750	0.14	5.5890	0.2180	3.90	3.66	5.16	1.31	25000	-0.03
104	665-652-B	3.3750	6.0000	1.6250	0.0562	1.6250	1.2500	0.14	6.2772	0.2812	4.02	3.74	5.55	1.43	39600	-0.31
105	841-832-B	3.3750	6.6250	2.1250	0.8750	2.2190	1.6250	0.14	6.9960	0.3750	4.09	3.82	6.10	1.95	65100	-0.73
106	759-752-B	3.5000	6.3750	1.8750	0.6870	1.9000	1.5000	0.14	6.6830	0.3120	4.17	3.90	5.91	1.71	52300	-0.47
107	6580-6535-B	3.5000	6.3750	2.1250	0.7500	2.1693	1.6875	0.14	6.7500	0.3125	4.29	4.01	6.10	1.46	51900	-0.52
108	850-832-B	3.5000	6.6250	2.1250	0.8750	2.1290	1.6250	0.14	6.9960	0.3750	4.17	3.94	6.10	1.95	65100	-0.73
109	760-752-B	3.5625	6.3750	1.8750	0.6870	1.9000	1.5000	0.14	6.6830	0.3120	4.21	3.98	5.91	1.71	52300	-0.47
110	47890-47825-B	3.6250	5.6250	1.3125	0.5937	1.3750	1.0313	0.14	5.8710	0.3125	4.21	3.98	5.59	1.30	31800	-0.04
111	47896-47825-B	3.7500	5.6250	1.3125	0.5937	1.3750	1.0313	0.14	5.8710	0.3125	4.33	4.06	5.59	1.30	31800	-0.04
112	594-592-B	3.7500	6.0000	1.5625	0.6250	1.4300	1.1875	0.14	6.2460	0.2500	4.33	4.09	5.67	1.32	34800	-0.10
113	594A-592-B	3.7500	6.0000	1.5625	0.6250	1.4300	1.1875	0.20	6.2460	0.2500	4.45	4.09	5.67	1.32	34800	-0.10
114	77375-77675-B	3.7500	6.7500	1.8750	0.6875	1.9000	1.5000	0.14	7.0584	0.3125	4.45	4.17	6.34	1.59	54300	-0.38
115	42381-42587-B	3.8125	5.8750	1.2500	0.5000	1.1406	0.9688	0.14	6.0898	0.2188	4.33	4.09	5.98	1.19	26400	0.12

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	T	T1	B	C	R 2/	D1	C1	d b	d a	D a	K factor	Basic dynamic load ratings (lb.)	a 3/
		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Max. shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
116	52400-52637-B	4.0000	6.3750	1.4375	0.6562	1.4219	1.0313	0.14	6.6210	0.2500	4.61	4.37	6.10	1.23	36100	-0.01
117	687-672-B	4.0000	6.6250	1.6250	0.7187	1.6250	1.1875	0.14	6.9030	0.2812	4.65	4.41	6.30	1.24	39800	-0.11
118	861-854-B	4.0000	7.5000	2.2500	0.8750	2.2650	1.7500	0.31	7.8710	0.3750	5.08	4.49	6.85	1.74	73200	-0.60
119	HH221449-HH221410-B	4.0000	7.5000	2.2500	0.8750	2.2650	1.8125	0.31	7.8710	0.4375	5.16	4.56	7.05	1.74	85200	-0.59
120	941-932-B	4.0000	8.3750	2.6250	0.9375	2.6250	2.1250	0.28	8.8085	0.4375	5.12	4.61	7.83	1.79	83300	-0.78
121	782-772-B	4.1250	7.1250	1.8750	0.6875	1.8900	1.5000	0.14	7.4330	0.3125	4.80	4.57	6.61	1.51	51400	-0.32
122	56425-56650-B	4.2500	6.5000	1.4375	0.6250	1.4375	1.0625	0.14	6.7460	0.2500	4.84	4.61	6.38	1.18	36600	0.08
123	64450-64700-B	4.5000	7.0000	0.9062	0.7188	1.6250	1.1875	0.14	7.2772	0.2813	5.16	4.92	6.85	1.13	44300	0.05
124	71450-71750-B	4.5000	7.5000	1.8750	0.8125	1.9375	1.3750	0.14	7.8080	0.3125	5.20	4.92	7.13	1.40	58300	-0.26
125	938-932-B	4.5000	8.3750	2.6250	0.9375	2.6250	2.1250	0.28	8.8085	0.4375	5.55	5.04	7.83	1.79	83300	-0.78
126	68462-68712-B	4.6250	7.1250	1.3750	0.6563	1.2500	1.0000	0.14	7.4022	0.2813	5.20	4.92	6.77	1.18	31800	-0.21
127	L624549-L624510-B	4.7500	6.3125	0.8438	0.3438	0.8438	0.6563	0.06	6.4647	0.1563	5.08	5.00	6.18	1.34	17300	0.33
128	795-792-B	4.7500	8.1250	1.8750	0.8125	1.8750	1.3750	0.13	8.4336	0.3125	5.47	5.28	7.80	1.27	57000	-0.07
129	48290-48220-B	5.0000	7.1875	1.5625	0.5000	1.5000	1.3125	0.14	7.4335	0.2500	5.55	5.31	6.97	1.91	43200	-0.22
130	67388-67322-B	5.0000	7.7500	1.8125	0.5938	1.8125	1.5000	0.14	8.0272	0.2813	5.67	5.43	7.52	1.70	56800	-0.25
131	95500-95925-B	5.0000	9.2500	2.5000	1.0000	2.5000	1.9375	0.25	9.6834	0.4375	6.06	5.59	8.54	1.58	89400	-0.55
132	799A-792-B	5.1250	8.1250	1.8750	0.8125	1.8750	1.3750	0.14	8.4336	0.3125	5.83	5.59	7.80	1.27	57000	-0.07
133	L327249-L327210-B	5.2500	6.9688	1.0000	0.3437	1.0313	0.8125	0.06	7.1210	0.1562	5.59	5.51	6.81	1.68	24000	0.16

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	T	T1	B	C	R 2/	D1	C1	d b	d a	D a	K factor	Basic dynamic load ratings (lb.)	a 3/
		Bore	Outside diameter	Bearing width	Standout	Cone width	Cup width	Max. shaft fillet radius	Flange		Backing shoulder diameters					Effective load center
	Cone-cup								Diameter	Width	Shaft		Housing		Radial	
134	67390-67322-B	5.2500	7.7500	1.8125	0.5938	1.8125	1.5000	0.14	8.0272	0.2813	5.87	5.63	7.52	1.70	56800	-0.25
135	95528-95925-B	5.2500	9.2500	2.5000	1.0000	2.5000	1.9375	0.19	9.6834	0.4375	6.18	5.83	8.54	1.58	89400	-0.55
136	48393-48320-B	5.3750	7.5000	1.5625	0.4688	1.5625	1.3125	0.14	7.7148	0.2188	5.94	5.67	7.32	1.82	43800	-0.16
137	74550-74850-B	5.5000	8.5000	1.8750	0.8125	1.8750	1.3750	0.14	8.8084	0.3125	6.22	5.94	8.23	1.20	57800	0.09
138	898-892-B	5.5000	9.0000	2.2500	0.8750	2.2500	1.7500	0.14	9.3710	0.3750	6.30	6.02	8.50	1.39	83300	-0.24
139	48685-48620-B	5.6250	7.8750	1.6250	0.5000	1.5625	1.3437	0.14	8.0898	0.2187	6.22	5.94	7.64	1.74	43500	-0.12
140	36690-36620-B	5.7500	7.6250	1.1250	0.3750	1.1250	0.9063	0.06	7.7772	0.1563	6.10	6.02	7.48	1.59	32000	0.19
141	82576-82950-B	5.7500	9.5000	2.2500	0.8750	2.2300	1.7500	0.14	9.8710	0.3750	6.54	6.30	8.90	1.32	82500	-0.14
142	HM231149-HM231115-B	5.8750	9.5000	2.2500	0.8750	2.2300	1.7500	0.14	9.8710	0.3750	6.57	6.42	8.82	1.83	86500	-0.45
143	L630349-L630310-B	6.0000	7.5625	0.9843	0.3925	0.9449	0.7480	0.08	7.7705	0.1562	6.38	6.22	7.44	1.40	13900	0.40
144	99600-99100-B	6.0000	10.0000	2.6250	1.1875	2.6250	1.8750	0.28	10.4320	0.4375	7.13	6.68	9.37	1.43	91400	-0.48

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3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing dimensions (and dynamic load ratings, see 3.4) shall conform to the requirements specified in table I for each of the bearing part numbers. The listed dimensions conform to the requirements specified in ABMA Standard 19.2, "Tapered Roller Bearings - Radial Inch Design". The bearing size shall be specified in the acquisition order (see 7.3(b)). For any unlisted bearing size codes, the associated dimensional and dynamic load rating requirements should also be specified in the acquisition order.

3.2 Materials

3.2.1 Cones (inner rings), cups (outer rings), and rollers. The bearing cones, cups, and rollers shall be made of case carburized or through-hardened steel produced in accordance with ASTM A 295/A 295M, "Standard Specification for High-Carbon Anti-Friction Bearing Steel", or ASTM A 534, "Standard Specification for Carburizing Steels for Anti-Friction Bearings". The steel shall show a fine fracture grain size in accordance with ASTM E 112, "Standard Test Methods for Determining Average Grain Size". Material hardness shall be no less than 58 HRC and no more than 64 HRC as defined in ASTM E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials".

3.2.2 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The cages shall be made from carbon steel (one piece stamped). Materials shall operate from -65 to 230 °F.

3.3 Tolerance class. The tolerance limits for bearings shall conform to tolerance class 4 as tabulated in ABMA Standard 19.2. Allowable tolerances for bearing components and assembled bearings are listed in tables II through VI.

3.4 Dynamic load rating. The bearing dynamic load rating shall conform to the requirements specified in table I for each bearing size code. The listed ratings conform to the requirements specified in ABMA Standard 11, "Load Ratings and Fatigue Life for Roller Bearings".

3.5 Effective load center. Dimension (a) in figure 1 locates a point on the cone axis, which is the center of pressure of all resisting forces set up by the rollers. All moments should be calculated from this point when determining bearing loading and shaft stresses. A plus value of (a) indicates that the center is outside the cone backface.

3.6 Lubrication. The bearings shall be furnished without lubrication.

3.7 Normal angle. A normal angle bearing has a contact angle between 10 and 19 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

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TABLE II. Cone bore tolerance.

Cone bore (d)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	6.0000	10	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE III. Cup diameter tolerance.

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE IV. Standout tolerance.

Standout (T1)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	4.0000	80	0
4.0000	6.0000	140	100

Note: Allowable tolerances are in 0.0001 inch.

TABLE V. Cup flange tolerance.

Cup flange OD (D1)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	12.0000	20	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE VI. Assembled bearing tolerance.

Assembled bearing maximum radial runout		
Cup OD (D)		Tolerance
Over	Incl.	
0	24.0	20

Note: Allowable tolerances are in 0.0001 inch.

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packaging, and marking. Unless otherwise specified in the acquisition order, the bearings shall be preserved, packaged, and marked in accordance with MIL-DTL-197, "Packaging of Bearings, Associated Parts and Subassemblies" (see 7.3(c)).

7. NOTES

7.1 Sources of documents.

7.1.1 FAR. Copies of the FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://acquisition.gov/comp/far/index.html/.

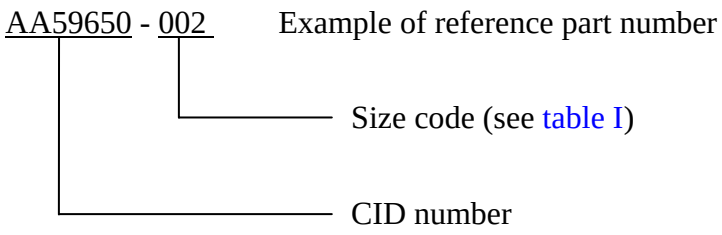
7.1.2 Military specifications. Copies of military specifications may be obtained from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies of military specifications may be obtained from http://assist.daps.dla.mil/.

7.1.3 ABMA standards. Copies of ABMA standards may be obtained from the American Bearing Manufacturers Association, 2025 M Street NW, Suite 800, Washington, DC 20036. Electronic copies of the ABMA standards may be obtained from http://www.abma-dc.org/.

7.1.4 ASTM standards. Copies of ASTM standards may be obtained from the ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. Electronic copies of the ASTM standards may be obtained from http://www.astm.org/.

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7.2 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



AA59650 - 002 indicates: bearing bore 0.5906 inches; outside diameter 1.3775 inches; width 0.4330 inches.

7.3 Ordering data. Acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1).
c. Preservation, packaging, and marking requirements (see 6.1).

7.4 Cross-reference information. Table VII relates the original specification slant sheets to the replacement CIDs.

TABLE VII. Cross-reference table.

FF-B-187B	Replacement CID	ABMA type
Specification sheet 1	A-A-59649	TS
Specification sheet 2	A-A-59650	TSF
Specification sheet 3	A-A-59651	TSS
Specification sheet 4	A-A-59652	TSSF
Specification sheet 5	A-A-59653	TDI & TDIS
Specification sheet 6	A-A-59654	TDO
Specification sheet 7	A-A-59655	TDOS
Specification sheet 8	A-A-59656	TNA (normal angle)
Specification sheet 9	A-A-59657	TNAS (steep angle)
Specification sheet 10	A-A-59658	TNASW
Specification sheet 11	A-A-59659	TNASWE

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7.5 Subject term (key word) listing.
bore
cone
cup
load
width

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - MC
Air Force - 99

Review Activities:
Navy - OS
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS4

(Project 3110-2006-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil/.

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